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Correct Answer

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Practice Exam - 1

Question 60 of 117



Which of the following set of test results is most likely to reveal rejection on endomyocardial biopsy in a 57-year-old female patient who is 10 months out from heart transplant, who feels well aside from some mild dyspnea going up steps or walking her dog 1 mile.

Which of the following sets of noninvasive surveillance test results is most likely to be associated with acute rejection on endomyocardial biopsy?

A. Allomap score of 15, dd-cfDNA of 0.04% and donor quantity score of 10

B. Allomap score of 35, dd-cfDNA of 0.04%, donor quantity score of 10

C. Allomap score of 38, dd-cfDNA of 0.68%, donor quantity score of 25

D. Allomap score of 20, dd-cfDNA of 0.22%, donor quantity score of 11

Commentary

Task (Choose one category: Diagnosis / Management / Test Selection / Interpretation / Prognosis): Diagnosis

Teaching Point: Noninvasive tests used for rejection surveillance should not be used in isolation. They complement each other and have individual limitations that alter NPV and PPV.

Correct Answer (Best Answer Choice): C.

Distractors (3–4 Plausible Incorrect Options):

1. The values for gene expression profiling (ie Allomap), %dd-cfDNA and the donor quantity score (value of dd-cfDNA) are all within normal. Each must be compared to prior values and used in context of patient signs/symptoms presently and in future follow-up. However, all above are within the reports range of normal.
2. In this set of tests, the gene expression profiling result is elevated (Allomap of 38). However, there are many things that can elevate this number. The %cd-ddDNA and DQS are both low, suggesting absence of notable graft injury. Taken together, this set of test is less likely to be associated with rejection than the set in answer C.

3. C is Correct!

4. In this set of tests, the %dd-cfDNA is mildly elevated, which could suggest clinically relevant graft injury. However, the DQS is <18.

There are many things that can impact the %dd-cfDNA (a ratio) that are not necessarily related to rejection. Taken together, this set of tests is less likely to be associated with rejection than the set in answer C.

Rationale – It is important to understand the individual weaknesses and benefits for gene expression profiling, %dd-cfDNA and the DQS. Gene expression profiling assesses up- and down-regulation of important genes associated with heart graft rejection. However, values are affected by degree of immunosuppression, concomitant illness, etc. The trend in the value is also critical. The %dd-cfDNA is a ratio of dd-cfDNA to recipient DNA. In rejection, this value increases due to elevation in dd-cfDNA more than the recipient DNA (which is mostly composed of DNA from recipient WBC turnover). However, other things can affect the %dd-cfDNA including BMI, infection, use of immunosuppression, malignancy etc. To help interpret abnormal %dd-cfDNA values, the DQS may be used. The DQS measures the quantity of donor cfDNA in the blood, ignoring the recipient cfDNA. If both the %dd-cfDNA (>0.15-0.2%) and DQS (>18) are above normal, this implies the donor cfDNA is likely elevated and myocardial injury is occurring. Answer C demonstrates an elevation of all tests and is more likely than A, B, and D to show rejection on biopsy.

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